

Crochet

User Guide

Version 004

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1 Introduction

Every Sixdof sensor is shipped to a customer with the necessary firmware already installed on the board during the production stage. Occasionally, a firmware update is required. For example, if the customer is getting a software update that requires a corresponding firmware update, or if a firmware bug was discovered at a later time. In these cases, it is desirable to be able to upgrade the firmware in the field. Additionally, the Sixdof sensor supports two modes for ip assignment, static ip and dynamic ip. It is desirable to be able to change the ip configuration in the field as well.

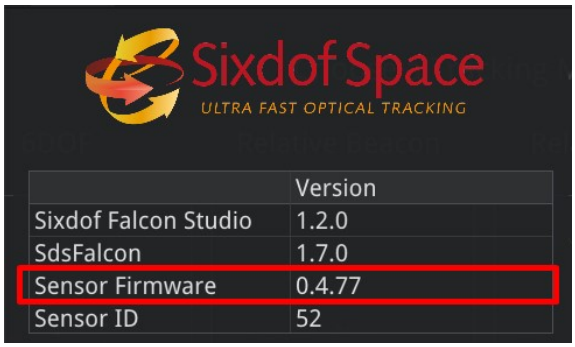
Crochet is a utility program that allows a user to upgrade the firmware on their sensor and change the ip connection settings. The process of burning a new version of firmware onto the sensor is called "boot loading". In this document we will describe how to use Crochet.

The Crochet program is only available on Windows.

2 Usage

2.1 Locate current firmware version

Before upgrading your sensors firmware version, it is important to know which is your current version. This way you can assess if a upgrade is necessary, and after the boot loading process, you can check the version again to confirm that the boot loading was successful. View the current firmware version by opening either Sixdof Falcon Studio, or Sixdof Cheetah Studio software. For assistance with either software, please consult the provided manuals. The sensor firmware is displayed in the "About" menu:



2.2 Bootload new Firmware version

Crochet software is only available on Windows, therefore the following steps can only be done on a Windows computer.

1. Ensure all other Sixdof software has been closed, all the necessary UDP ports must not be in use for Crochet to work correctly.
2. Locate the new firmware file provided by Sixdof. This file contains the firmware that will be burned onto the board. The file name will look like: "kipa_0_4_95_with_bl_0_1_16.hex".
3. Open a command prompt in "Administrator" mode.
4. Run the **crochet.exe** program from the Administrator command prompt and follow the instructions as they are being displayed. The instructions are listed below as well, with example responses in red :
   - Enter the name of the network the sensor is connected to: Ethernet
   - Enter the sensor id: 100
   - Do you want to bootload new firmware? [y/n] y
   - Enter full file path to the firmware file provided by Sixdof: `C:\\Users\\User\\Downloads\\kipa_0_4_95_with_bl_0_1_16.hex`
   - Do you want to set a new static ip address? [y/n] n
   - Do you want to revert to the default ip behavior (dynamic) [y/n] n
   - Restart the sensor and press Enter to continue... Press `Enter` within the first 3 seconds of restarting.

5. Once you receive a message that says "Burning Firmware complete", you can restart the board by disconnecting the power to put it back into regular operation mode.

6. It is recommended to check the firmware version to confirm that the changes took hold. See section 2.1 for instructions on how to check the version.

2.3 Change IP Behavior

The default behavior of the Sixdof sensor is that it is assigned an ip address on runtime, according to the ip address of the network it is connected to. There are some cases where this behavior is not desired and it is better to assign the sensor a fixed ip address. Note; these features are only supported by firmware version 0.4.95 and higher. If your sensor currently has a lower version of firmware, you should first bootload firmware version 0.4.95 onto the sensor and then continue with the following instructions. Below are the instructions on using Crochet to assign a sensor with a fixed ip address:

1. Ensure all other Sixdof software has been closed, all the necessary UDP ports must not be in use for Crochet to work correctly.

2. Identify the desired ip address of the sensor and the ip address of the host.

3. Open a command prompt in "Administrator" mode.

4. Run the **crochet.exe** program from the Administrator command prompt and follow the instructions as they are being displayed. The instructions are listed below as well, with example responses in red :
   - Enter the name of the network the sensor is connected to: Ethernet
   - Enter the sensor id: 100
   - Do you want to bootload new firmware? [y/n] n
   - Do you want to set a new static ip address? [y/n] y
   - Enter the desired ip address for the SENSOR: 192.168.0.25
   - Enter the desired ip address for the HOST computer: 192.168.0.10
   - Do you want to revert to the default ip behavior (dynamic) [y/n] n
   - Restart the sensor and press Enter to continue... Press `Enter` within the first 3 seconds of restarting.

5. Once you receive a message that says "Burning Static IP complete", you can restart the board by disconnecting the power to put it back into regular operation mode.

In some cases it is desirable to switch back from a fixed ip address to the sensors default behavior. Below are the instructions on using Crochet to go back to the default behavior:

1. Ensure all other Sixdof software has been closed, all the necessary UDP ports must not be in use for Crochet to work correctly.

2. Open a command prompt in "Administrator" mode.

3. Run the **crochet.exe** program from the Administrator command prompt and follow the instructions as they are being displayed. The instructions are listed below as well, with example responses in red :
   - Enter the name of the network the sensor is connected to: Ethernet
   - Enter the sensor id: 100

- Do you want to bootload new firmware? [y/n] n
- Do you want to set a new static ip address? [y/n] n
- Do you want to revert to the default ip behavior (dynamic) [y/n] y
- Restart the sensor and press Enter to continue... Press `Enter` within the first 3 seconds of restarting.

4. Once you receive a message that says "Burning Dynamic IP complete", you can restart the board by disconnecting the power to put it back into regular operation mode.

2.4 Command line parameters

Instead of responding to prompts, the user inputs can be passed in via command line parameters. Descriptions of the command line parameters are shown below:

Short name	Long name	Description
-d	–debug_network	Shows network debug information in terminal window
-f	–firmware	Full path to the firmware file provided by Sixdof
-n	–network	Network name that the sensor is connected to
-s	–sensor	Sensor id, this is the number on the label of the sensor, leave out leading zeros
	–ip_sensor	New IP address of the sensor
	–ip_host	New IP address of the host
	–ip_default	Sets the static ip behaviour back to default (dynamic)
-q	–quick_start	Quick start Crochet, skip all user prompts and only use data provided by the command line
-h	–help	Shows the help menu